

An Early Stage Determination of Colon Cancer Through Deep Neural Network

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Abstract— Identification and prediction of Colon cancer have always been difficult and essential tasks for medical professionals. Hospitals are providing costly therapies and operations to treat the illness. So, diagnosing colon cancer at an early stage will be helpful to patients around the world. Currently, the recent improvements in medical technologies, Deep Learning techniques exhibit a major role in predicting the development of colon cancer. In this research work, the risk factors that produce colon cancer-causing genes are selected using feature selection methods such as Fisher Score Univariate Filter (FSUF) and ReliefF Multiclass Filter (RFMF), which are applied separately and integrate the selected prominent genes to minimize the dimensionality of the data. To predict colon cancer, Deep Neural Network (DNN) classifier is utilized on the reduced data and computes the classifier's performance using evaluation metrics. Finally, the outcomes of experiments are compared with two machine learning algorithms, like Independence Bayes (IB) classification model and the k-nearest neighbors (k-NN) classifier. The experimental outcomes demonstrate that the research model significantly reduces the dimension of data space and produced an accurate prediction of colon cancer when compared with other classifiers.

Keywords— Colon cancer, Deep Learning, Feature Selection method, Fisher Score Univariate Filter, ReliefF Multiclass Filter, Machine Learning

I. INTRODUCTION

Colon cancer is a specific type of cancer that develops in the large intestine part and represents one of the most deadly diseases among all age groups. But the progression of the disease can be reduced by proper treatment during the early stages. Global Cancer Observatory (GCO) Database 2022 examined new 1,931,590 colon cancer patients and 935,173 colon related deaths. In the human body, colon cancer develops as a result of variations in DNA within the cells that may have an impact on oncogenes and tumor suppressor genes, triggering numerous mutations in genes that may result in the colon. High-dimensional gene data based on microarray technology are primarily utilized in the analysis and detection of colon disease.[1]. In the cancer dataset, the values of the expression of genes are arranged in a matrix form, where the rows relate to the patient sample instance and the columns to different genes[2]. The gene dataset contains an excessive number of genes and relatively few samples, plenty of genes are redundant and noisy. The values in the dataset are real numbers and describe the levels of gene expression. Due to the curse of high dimensionality, attribute subset selection techniques are employed to select

significant genes that cause colon cancer and to improve the accuracy of the classification task.

In microarray gene expression data, the selection of features is represented as gene selection and refers to the process of selecting a significant number of relative and important genes. Embedded, filter, wrapper, hybrid, and ensemble methods are commonly used attribute selection techniques[3]. Embedded methods estimate the attributes that have the highest efficiency on the accuracy of the model. Filter methods are used to estimate the feature relationship measures such as correlation, redundancy and relevance which are executed independently to the prediction model to select the subset of features[4]. Wrapper methods apply a greedy approach and check all possible combinations of features to identify an optimal subset of features. Hybrid methods combine embedded, filter, and wrapper methods to find important features in the dataset. Ensemble methods aggregate the outputs of multiple feature subsets and produced a prominent set of features. Currently, ML and DL methods are primarily used in the bioinformatics domain to identify the genes of oncological patients.[5]

The key objective of this study is to design a computational framework for effectively identifying colon cancer at an early stage. The designed model integrates FSUF and RFMF gene selection methods for efficiently identifying the key subset of genes from colon cancer microarray data. Also, identified weak genes in the data set that have a low impact on the detection of the disease. The selected prominent features are incorporated into the Deep Neural Network for accurate diagnoses.

The remaining part of the research paper is structured as follows: Section 2 outlined the important existing methods of colon gene prediction. Section 3, explained the selection of genes and the classification algorithm. Section 4 elaborates on the system that is being proposed. Section 5 provides an experimental research analysis that summarizes the performance parameters used for different algorithms. Section 6 summarizes the conclusions of these research findings.

II. RELATED WORK

In recent Machine and Deep learning literature, a lot of research efforts have focused on using a microarray-based selection of genes for cancer disease. A subset of prominent genes is selected from the collection of original genes through the process of gene selection. Md. Akizur Rahman et al. [6] explored an interesting technique to classify colon

cancer microarray data. Best first search feature selection technique was applied to reduce the number of attributes and selected only a subset of relevant features. The proposed algorithm was implemented in Artificial Neural Networks (ANNs) for colon cancer classification and achieved 98.40% performance accuracy using MATLAB tools. Dhana Sri Devi et al [7] proposed an ensemble model to determine the optimal set of features using entropy-based IG and GW optimization algorithms. RF classification was applied to increase the accuracy percentage. Soumiya Hamena et al.[8] proposed multi class cancer classification using deep learning techniques. The model used two activation functions, namely Rectified Linear Units and Sigmoid functions in the hidden layers to process the cancer gene data and evaluate the performance based on loss percentage and accuracy value.

Pyingkodi M et al.[9] designed a heuristic search technique to select highly relevant genes. The GALF method was implemented for the identification of cancer in genomics data. The experimental output indicates that the research method is appropriate for the selection of genes, removing irrelevant and redundant genes present in the dataset and also increasing the classification accuracy rate. Hanafi Abdullah Gusman et al. [10] employed the DWTransform reduction method to select the informative genes. SVM and K-NN models are applied to produce an accurate prediction of disease. Lin Sun et al.[11] formed a system for determining tumor through the ReliefF and ACO techniques. The research method can significantly decrease the attribute size and increase the performance measure.

According to the literature, a highly specific methodology is required for classifying colon cancer gene data. The present research is focused on achieving the desired outcomes.

III. MATERIALS AND METHODS

The research methodology attempts to recognize cancer genes with genomics data and improve the process of data classification. The microarray gene-based high-dimensional data represents a significant challenge to the classification of genes. Selecting prominent genes from thousands of genes that can directly contribute to the symptoms of the disease is an important problem[12]. Feature selection techniques are essential for reducing attribute size and eliminating irrelevant data. Due to the high degree of dimensionality, the DNN classification model is employed to accurately predict colon cancer.

A. Fisher Score Univariate Filter(FSUF)

Fisher Score Univariate filter is one of the famous supervised feature selection techniques and aims to measure the relevance between predictors and the output variable in order to determine the important features from the data. The score is obtained based on statistical measures such as the mean and variance values of the attribute[13]. Heuristic method is utilized to compute a score for each attribute independently and select the attributes with the highest score value. This method also eliminated the risk of overfitting. The approach cannot handle redundant features and selected

suboptimal features. Algorithm-1 represents the steps of the FSUF method.

```
//Algorithm-1(Gene Selection method-FSUF)
Input:  $X_1, X_2, \dots, X_n$  are predictors and Y output variable
Output: Assign score values to all the attributes.
n: Number of Training instances
n+: Number of instances of class0
n-: Number of instances of class1
k: Number of classes
For i:=1 to n do begin
   $F1[i] = (\overline{X_i^{(+)}} - \overline{X_i})^2 + (\overline{X_i^{(-)}} - \overline{X_i})^2$ 
   $F2[i] =$ 
   $(1/n-1) * \sum_{k=1}^{n-1} (X_{k,i}^{(+)} - \overline{X_i^{(+)}})^2 + (1/n-1) * \sum_{k=1}^{n-1} (X_{k,i}^{(-)} - \overline{X_i^{(-)}})^2$ 
   $F[i] = F1[i]/F2[i]$ 
End For
return score of each feature.
```

B. ReliefF Multiclass Filter (RFMF)

The RFMF is a fast and powerful filter approach to feature weighing, and the output vector is a weight value between -1 and 1 for each attribute. To identify conditional relationships between attributes and generate an integrated view of the attribute estimation. The RFMF algorithm selects an instance randomly and calculates the nearest Hit and Miss values. Feature weight is calculated based on the differences between the pairs of nearest neighbor instances. The weight value decreases when neighboring samples correspond to the same class and increases when instances are assigned to different classes[14]. The method is computationally less expensive and more suitable for high-dimensional datasets[15]. Algorithm-2 shows the pseudo-code of RFMF gene selection method.

```
//Algorithm -2 (Gene Selection method-RFMF)
Input:  $X = \{X_1, X_2, \dots, X_n\}$  are independent attributes and D dependent attribute
Output: Assign weight values to all the attributes.
m: Number of Training instances
A: Number of attributes
Set weights of all attributes  $W[A] := 0.0$ 
For i:=1 to m do begin
  Randomly select an instance of  $R_i$ 
  Find the nearest value of hit H and nearest value of miss M.
  For j=1 to A do
     $W[j] < W[j] - \text{Diff}(j, R_k, H)/m + \text{Diff}(j, R_k, M)/m$ 
  End For
End For
Return W
```

C. Deep Neural Networks Model

Deep Neural Network is based on ANN with a layered network that consists of multiple layers between the Input and Output layers. The nodes initial weights are assigned randomly, and calculate the error rate by adjusting the weight values to get an accurate outcome[16]. The stopping condition is based on the epoch count or the expected output is obtained. The network will require more time to train the model if the number of hidden layers is increased [17]. DNN classification model is primarily utilized with gene data due to the high dimensionality to classify the disease

with high accuracy value[18]. Algorithm-3 shows the pseudo-code of the DNN gene classification method.

```
//Algorithm-3(Deep Neural Network classification model)
1.Read N number of nodes for the input layer.
2.Initialize the number of hidden layers that need to be
trained with data.
3. Initialize the learning rate and bias values for every node.
4. The weight will be assigned randomly in forward
propagation.
5. Define the Hidden layer activation function.
  Apply Rectified Linear Unit(ReLU):
     $f(x)=\max(0,x)$  activation function
6. Initialize the Output layer with the activation function
  Apply Sigmoid function :
     $f(x)=1/(1+e^{-x})$  activation function
7. Define the number of epoch to process the value from the
input node.
8. Train the network till the defined number of epoch is
reached.
9. Test the network to classify the data.
10.Calculate the performance of the model using evaluation
metrics.
```

IV. RESEARCH METHODOLOGY

This section explored the proposed FSUF-RFMF-DNN system for the identification of the most essential genes with predictive value. Fig.1 presents the structure of the research model.

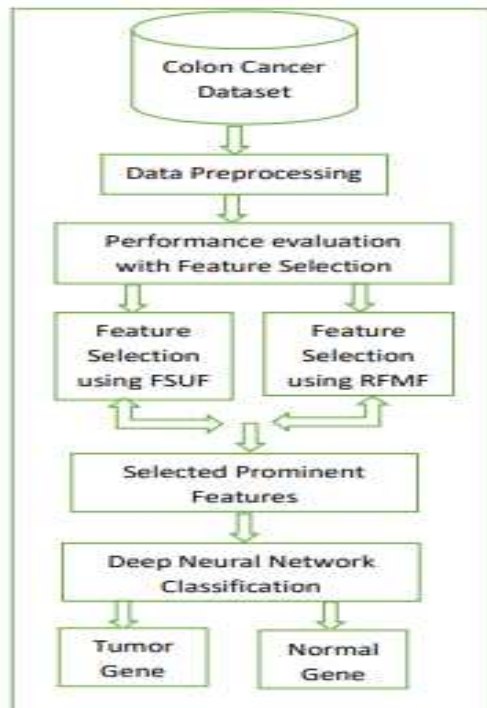


Fig.1 Structure of the Research Model

The procedure consists of three stages, which are Data preprocessing , gene selection, and model classification. Algorithm-4 presents the key steps of the proposed system.

```
//Algorithm-4(FSUF-RFMF-DNN)
Input:- A n-dimensional DNA colon dataset
DS= {X1, X2,...Xn,Y}
      where X1, X2,...Xn denotes independent genes and
      Y represents dependent gene
Output: Prediction of Genes.
k: Number of selected features using FSUF and RFMF
method
Scaled_data: Normalized data using Z-Score Normalization
FS_features: FSUF selected genes
RF_features:RFMF selected genes
PF_features: Prominent set of genes
DS : Colon Cancer Dataset
/* Apply Normalization using Z-Score
Scaled_data<-Normalization(DS)
/* Apply FSUF Filter Method
FS_features<-FSUF(k,Scaled_data)
/* Apply RFMF Filter Method
RF_features<-RFMF(k,Scaled_data)
/* Integrate FSUF and RFMF
PF_features<-(FS_features) Union (RF_features)
Split dataset using Training and Testing set.
Prediction_models<-Classification_Model(Xtrain,Xtest)
Return all generated prediction models with their evaluation
performances.

Function Classification Model(Training_set,Test_set):
  List_Classifiers=[DNN,NB,KNN]
  Model<-Train Classifier on the Training set
  Test the Model on the Testing set
  Calculate Performance metrics
    (Accuracy,Precision,Recall,F1-Score)
  Return All obtained Models with their performances.
```

In the first stage, to collect the colon cancer dataset from the NCBI repository. The cancer dataset consists of 2000 colon cancer genes and 62 patient samples. In the total samples, 40 samples were captured from Tumor cells and 22 samples from Normal cells. The dataset supports binary classification models, with output values of 0 for (Abnormal)-Tumor and 1 for Normal gene. Data preprocessing stage helps to measure the central tendency of the data by checking the missing values and Null values present in the data and also extracting useful patterns from the data. In this stage to improve the prediction performance, normalize the cancer dataset prior to feeding gene selection and classification. In this process, each gene is normalized based on the Z-Score normalization technique After pre-processing, gene selection methods such as FSUF and RFMF are applied to remove redundant and noisy genes in the dataset and select the prominent genes causing cancer.

The selected 822 prominent genes are applied in the Deep neural network classification model, which includes an Input layer, five Hidden layers with 30 neurons, and an Output layer corresponding to the Normal gene or Tumor gene classes. The network is trained using Adam optimizer at a learning rate of 0.001. The end of the training is controlled by an early stop with a maximum of 30 epochs. The activation functions, such as Relu and Sigmoid, were applied in the hidden and output layers, respectively. A loss function

such as Sparse categorical cross-entropy is applied to measure the loss value.

The functionality of a model is calculated using a confusion matrix, which helps determine the correct and incorrect samples present in the system. The parameters used in the research study are given in Table I.

TABLE I. PARAMETERS USED IN THE RESEARCH METHODOLOGY

Accuracy	Accuracy= $(TP+TN)/(TP+FN+TN+FP)$	TP: True Positive TN: True Negative FP: False Positive FN: False Negative
Precision	$PC_0 = (TP \text{ of Class } 0)/(TP+FP)$ $PC_1 = (TP \text{ of Class } 1)/(TP+FP)$ Macro Average Precision= $(PC_0+PC_1)/2$ Weighted Average Precision= $((PC_0 * C_0) + (PC_1 * C_1))/(C_0+C_1)$	PC_0 : Precision of Class 0 PC_1 : Precision of Class 1 C_0 : Total Samples of Class 0 C_1 : Total Samples of Class 1
Recall	$RC_0 = (TP \text{ of Class } 0)/(TP+FN)$ $RC_1 = (TP \text{ of Class } 1)/(TP+FN)$ Macro Average Recall= $(RC_0+RC_1)/2$ Weighted Average Recall= $((RC_0 * C_0) + (RC_1 * C_1))/(C_0+C_1)$	RC_0 : Recall of Class 0 RC_1 : Recall of Class 1
F1-Score	$F1S_0 = 2 * (PC_0 * RC_0) / (PC_0 + RC_0)$ $F1S_1 = 2 * (PC_1 * RC_1) / (PC_1 + RC_1)$ MacroAverage F1Score= $(F1S_0+F1S_1)/2$ Weighted Average F1-Score= $((F1S_0 * C_0) + (F1S_1 * C_1))/(C_0+C_1)$	$F1S_0$: F1-Score of Class 0 $F1S_1$: F1-Score of Class 1

Macro average precision is determined as the arithmetic mean of each class precision. Weighted average precision measures, the total sample count multiplied by the precision of each of the classes, and dividing the result by the total sample count. Similarly, Macro average recall, F1-Score, and Weighted average recall, F1-Score are implemented.

V. EXPERIMENTAL ANALYSIS

The research methodology has been developed and implemented in Python Google Colob Notebook. This section comprises several experiments with the colon cancer dataset. In the study, the network was trained with 70 percent of the data and 30 percent for testing purposes.

Table-II indicates the Accuracy and Loss measure of Training and Validation data of FSUF-RFMF-DNN model and predicted an accuracy rate of 94.44 % using the test data.

Fig-2 indicates the performance accuracy measure of the research work. The system utilized 30 epochs to train the data. In epoch-5 the Training accuracy rate is 84% and the Validation accuracy rate is 61.11% respectively. In epoch-10, the Training and Validation accuracy values increased to 92% and 66.67% respectively. Similarly, in epoch-15,20,25 and 30 the Training and Validation accuracy values also increased.

Fig.3 depicts the performance Loss measure of proposed approach. In epoch-5 the Training loss rate is 42.4% and Validation loss rate is 56.05%, In epoch-10 the Training loss rate is decreased to 30.04% and Validation loss is 50.75%. Similarly, epoch-15,20,25 and 30 the Training loss and Validation loss rate is decreased. 31.27% of loss value obtained to test the data in the model. The

accuracy measures for the FSUF-RFMF-DNN, KNN and Naïve Bayes models are compared in Fig.4. FSUF-RFMF-DNN method attained a high accuracy rate of 94.44% compared to other methods.

Table-III depicts the comparison metrics of FSUF-RFMF-DNN model and other models. Fig.5 illustrates the performance measure of Precision and depicts that the proposed method outperformed high precision rate in terms of Macro average precision of 83% and Weighted average precision of 93% in comparison with other models such as K-NN and NB. Fig.6 shows the comparison measure of Recall and demonstrates that the research approach produced 93% for Macro average Recall and 89% for Weighted average Recall. K-NN performed 90% and 84% for Macro and Weighted average Recall values respectively. Fig.7 demonstrates the performance measure of the F1-Score. NB classifier produced a low Macro average score of 68% and Weighted average score of 76% compared to the proposed and K-NN models. The final results showed that the proposed technique FSUF-RFMF-DNN provides fast and effective solutions in the analysis of high dimensional colon cancer data.

TABLE II. ACCURACY AND LOSS MEASURES OF FSUF-RFMF-DNN METHOD

Epoch	Training Accuracy(%)	Validation Accuracy(%)	Training Loss(%)	Validation Loss(%)
5	84	61.11	42.4	56.05
10	92	66.67	30.04	50.75
15	96	83.33	15.17	41.20
20	100	88.89	8.27	35.81
25	100	94.44	2.17	31.51

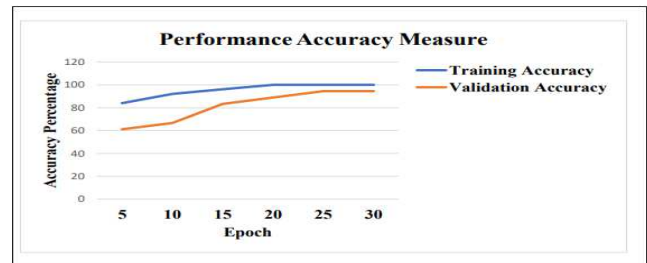


Fig. 2 Plot of Accuracy versus Epoch for FSUF-RFMF-DNN method



Fig. 3. Plot of Loss versus Epoch for FSUF-RFMF-DNN method

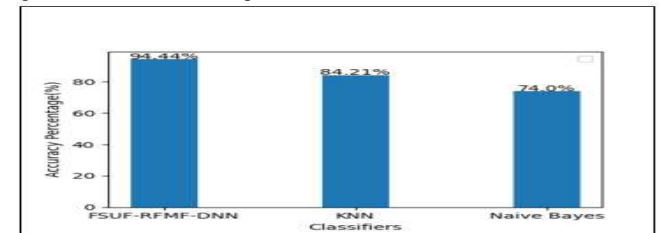


Fig. 4. Accuracy measure for FSUF-RFMF-DNN, KNN, NB classifiers.

TABLE III. COMPARISON METRICS OF FSUF-RFMF-DNN,KNN, NB CLASSIFIERS

Model	Proposed Method FSUF-RFMF-DNN			KNN			Naive Bayes		
Metrics(%)	Precision	Recall	F1-Score	Precision	Recall	F1-Score	Precision	Recall	F1-Score
Class 0 (Tumor)	100	87	93	100	80	89	92	73	81
Class 1 (Normal)	67	100	80	57	100	73	43	75	55
Macro Average	83	93	86	79	90	81	67	74	68
Weighted Average	93	89	90	91	84	85	81	74	76

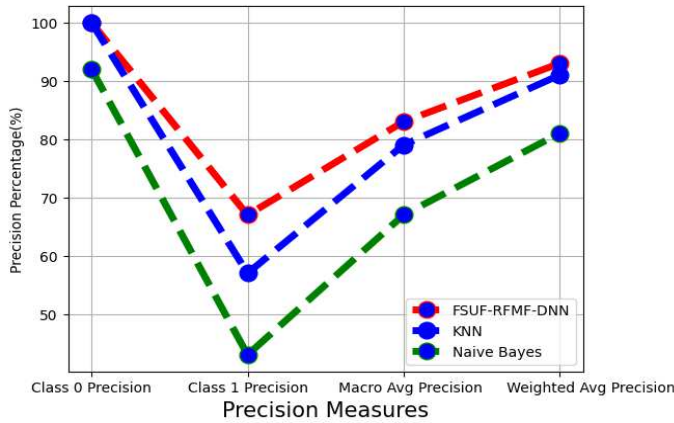


Fig. 5. Precision metrics for FSUF-RFMF-DNN, KNN, and NB methods.

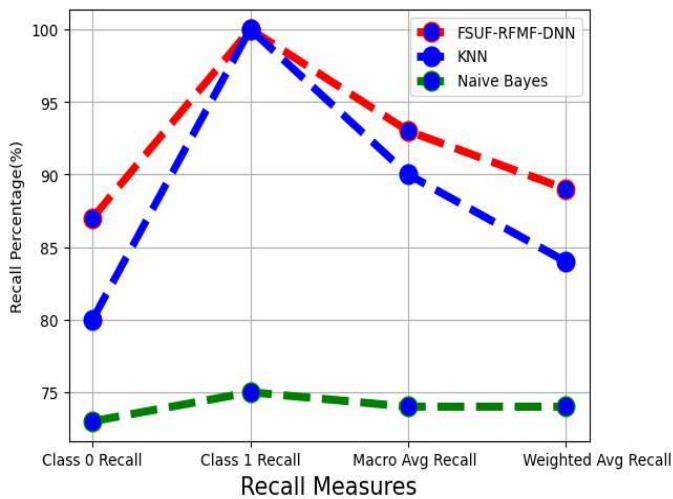


Fig. 6. Recall metrics for FSUF-RFMF-DNN, KNN, and NB methods.

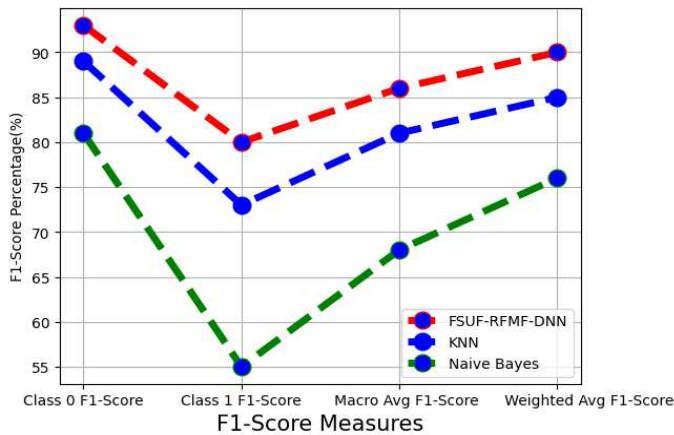


Fig 7. F1-Score metrics for FSUF-RFMF-DNN, KNN, and NB methods.

Many investigators have explored different techniques for the determination of colon genes. Table-IV shows the computational efficiency of various algorithms. The outcomes of the research study compared with other investigated methods are shown in Fig.8

TABLE IV: COMPUTATIONAL EFFICIENCY OF THE SURVEYED METHOD

Reference	Methodology	Accuracy(%)
[19]	Genetic Algorithm+Decision Tree(c4.5)	88.71%
[20]	Fisher Score+ Kmeans+Harmony Search	93.36%
[21]	Fisher Score+ Chemelon Algorithm	85.48%
[22]	MRMR+COA-HS+SVM	93.23%
[23]	K-Means+ReliefF+RandomForest	85.87%
	FSUF-RFMF-DNN(Proposed)	94.44%

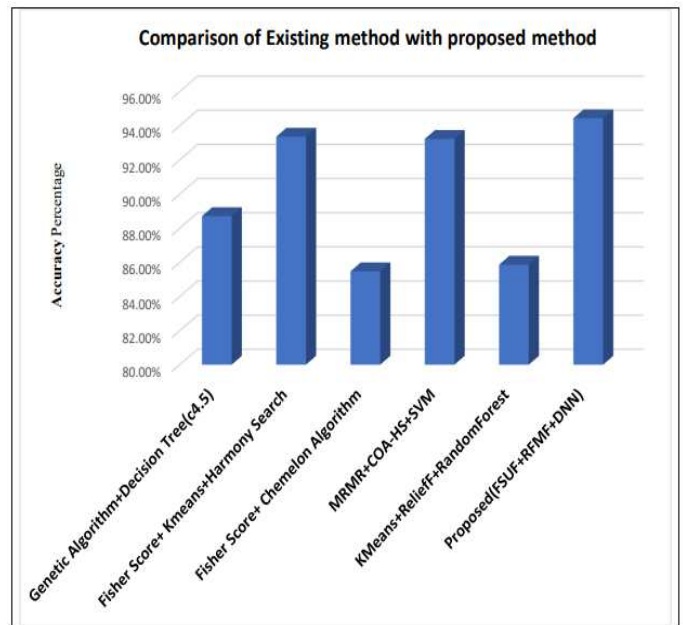


Fig. 8 Comparison of surveyed methods

VI. CONCLUSIONS

The main key to finding the disease at an early stage is obtaining an effective treatment for colon cancer. This research work has developed a new technique for the early identification of colon cancer. The proposed technique performs data preprocessing, which includes checking the central tendency of the data, extracting useful patterns, and normalizing the gene values using z-score normalization. After preprocessing, essential genes are identified through FSUF and RFMF methods. Also, identified weak genes in the data set that have a low impact on the detection of the disease and minimized the high dimensionality of the biomedical colon cancer data. The prominent genes are utilized to categorize diseases with a Deep Neural Network

classifier. The proposed method eliminates the similar and noisy genes in the high-dimensional microarray gene expression colon dataset and produced high classification accuracy. Furthermore, comparisons are performed with other models. In the future, gene selection and classification performance can be derived from the design of clustering and other deep learning techniques.

REFERENCES

- [1] Shafi, A. S. M., MM Imran Molla, Julakha Jahan Jui, and Mohammad Motiur Rahman. "Detection of colon cancer based on microarray dataset using machine learning as a feature selection and classification techniques." *SN Applied Sciences* 2 (2020): 1-8.
- [2] Islam, Ashraful, Mohammad Masudur Rahman, Eshtiaq Ahmed, Faisal Arafat, and Md Fazle Rabby. "Adaptive feature selection and classification of colon cancer from gene expression data: an ensemble learning approach." In *Proceedings of the international conference on computing advancements*, pp. 1-7. 2020.
- [3] Ge, Chenyu. "Selection of potential cancer biomarkers based on feature selection method." In *Third International Conference on Intelligent Computing and Human-Computer Interaction (ICHCI 2022)*, vol. 12509, pp. 227-236. SPIE, 2023.
- [4] Lee, Junghye, In Young Choi, and Chi-Hyuck Jun. "An efficient multivariate feature ranking method for gene selection in high-dimensional microarray data." *Expert Systems with Applications* 166 (2021): 113971.
- [5] Adornetto, Carlo, and Gianluigi Greco. "A New Deep Learning and XAI-Based Algorithm for Features Selection in Genomics." *arXiv preprint arXiv:2303.2023*: (2023): 16914
- [6] Rahman, Md Akizur, and Ravie Chandren Muniyandi. "Feature selection from colon cancer dataset for cancer classification using artificial neural network." *Int. J. Adv. Sci. Eng. Inf. Technol* 8, no. 4-2 (2018): 1387-1393.
- [7] Sree Devi, K. Dhana, P. Karthikeyan, Usha Moorthy, K. Deeba, V. Maheshwari, and Shaikh Muhammad Allayear. "Tumor Detection on Microarray Data Using Grey Wolf Optimization with Gain Information." *Mathematical Problems in Engineering* 2022 (2022).
- [8] Hamena, Soumiya, and Souham Meshoul. "Multi-class classification of gene expression data using deep learning for cancer prediction." *Int J Mach Learn Comput* 8, no. 5 (2018): 454-459.
- [9] Pyngkodi, M., and R. Thangarajan. "Informative gene selection for cancer classification with microarray data using a metaheuristic framework." *Asian Pacific journal of cancer prevention: APJCP* 19, no. 2 (2018): 561.
- [10] Gusman, Hanafi Abdullah, and Widi Astuti. "Microarray Data Classification to Detect Cancer Cells by Using Discrete Wavelet Transform and Combining Classifiers Methods." In *2020 International Conference on Data Science and Its Applications (ICoDSA)*, pp. 1-7. IEEE, 2020.
- [11] Sun, Lin, Xianglin Kong, Jiucheng Xu, Zhan'ao Xue, Ruibing Zhai, and Shiguang Zhang. "A hybrid gene selection method based on ReliefF and ant colony optimization algorithm for tumor classification." *Scientific reports* 9, no. 1 (2019): 8978.
- [12] Pashaei, Elnaz, Mustafa Ozen, and Nizamettin Aydin. "Gene selection and classification approach for microarray data based on Random Forest Ranking and BBHA." In *2016 IEEE-EMBS International Conference on Biomedical and Health Informatics (BHI)*, pp. 308-311. IEEE, 2016.
- [13] Gu, Quanquan, Zhenhui Li, and Jiawei Han. "Generalized fisher score for feature selection." *arXiv preprint arXiv:1202.3725* (2012).
- [14] Lopez, Nahum Cueto, Maria Teresa García-Ordás, Facundo Vitelli-Storelli, Pablo Fernández-Navarro, Camilo Palazuelos, and Rocío Alaiz-Rodríguez. "Evaluation of feature selection techniques for breast cancer risk prediction." *International Journal of Environmental Research and Public Health* 18, no. 20 (2021): 10670.
- [15] Haq, Amin Ul, Jian Ping Li, Abdus Saboor, Jalaluddin Khan, Samad Wali, Sultan Ahmad, Amjad Ali, Ghufraan Ahmad Khan, and Wang Zhou. "Detection of breast cancer through clinical data using supervised and unsupervised feature selection techniques." *IEEE Access* 9 (2021): 22090-22105.
- [16] Li, Kexuan, Fangfang Wang, Lingli Yang, and Ruiqi Liu. "Deep feature screening: Feature selection for ultra high-dimensional data via deep neural networks." *Neurocomputing* 538 (2023): 126186.
- [17] Karthik, S., R. Srinivasa Perumal, and P. V. S. S. R. Chandra Mouli. "Breast cancer classification using deep neural networks." *Knowledge Computing and Its Applications: Knowledge Manipulation and Processing Techniques: Volume 1* (2018): 227-241.
- [18] Gupta, Surbhi, Manoj K. Gupta, Mohammad Shabaz, and Ashutosh Sharma. "Deep learning techniques for cancer classification using microarray gene expression data." *Frontiers in Physiology* (2022): 2002.
- [19] Pavithra, D., and B. Lakshmanan. "Feature selection and classification in gene expression cancer data." In *2017 International Conference on Computational Intelligence in Data Science (ICCIDS)*, pp. 1-6. IEEE, 2017.
- [20] Bae, Jin Hee, Minwoo Kim, J. S. Lim, and Zong Woo Geem. "Feature selection for colon cancer detection using k-means clustering and modified harmony search algorithm." *Mathematics* 9, no. 5 (2021): 570.
- [21] Xie, Juanying, Yuchen Wang, and Zhaozhong Wu. "Colon cancer data analysis by chameleon algorithm." *Health Information Science and Systems* 7 (2019): 1-8.
- [22] Elyasigomari, Vahid, D. A. Lee, Hazel RC Screen, and M. Hasan Shaheed. "Development of a two-stage gene selection method that incorporates a novel hybrid approach using the cuckoo optimization algorithm and harmony search for cancer classification." *Journal of biomedical informatics* 67 (2017): 11-20.
- [23] Aydadenta, Husna, and Adiwijaya Adiwijaya. "A clustering approach for feature selection in microarray data classification using random forest." *Journal of Information Processing Systems* 14, no. 5 (2018): 1167-1175.